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Neural symphony of flow experience: Evidence for high-dimensional metastable dynamics

Abdelrahman B.M. Eldaly [a,b], Kris Zhangguang Kang [c], Fiona Fui-Hoon Nah [d], Leanne Lai-Hang Chan [a], Keng Leng Siau [d], Xiao Fan Liu [e], Richard Huskey [f], Langtao Chen [g], Tejaswini Yelamanchili [h], René Weber [i,j,*]

[a] Department of Electrical Engineering, City University of Hong Kong, Kowloon, Hong Kong SAR, China
[b] Electrical Engineering Department, Faculty of Engineering, Minia University, Minia, Egypt
[c] Department of Information Systems, City University of Hong Kong, Kowloon, Hong Kong SAR, China
[d] School of Computing and Information Systems, Singapore Management University, Singapore
[e] Department of Media and Communication, City University of Hong Kong, Kowloon, Hong Kong SAR, China
[f] Department of Communication, University of California, Davis, Davis, CA, USA
[g] Collins College of Business, The University of Tulsa, Tulsa, OK, USA
[h] Department of Business and Information Technology, Missouri University of Science and Technology, Rolla, MO, USA
[i] Media Neuroscience Lab, Department of Communication, University of California, Santa Barbara, Santa Barbara, CA, USA
[j] Department of Psychological and Brain Sciences, University of California, Santa Barbara, Santa Barbara, CA, USA

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ABSTRACT

Flow, an optimal experience characterized by deep immersion and engagement in an activity, has been extensively studied in behavioral research. However, its neural dynamic mechanism remains poorly understood. In a within-subject video gaming experiment, we captured neural activity underlying flow, boredom, and anxiety using a 64-channel electroencephalogram (EEG) system. Compared to boredom and anxiety, flow exhibits the highest global functional connectivity, metastability, and dimensionality of dynamic functional connectivity patterns, suggesting that flow is a highly adaptable process that is supported by high-dimensional neural dynamics. Unlike previous studies that focused on identifying static or localized brain activity, we examine the neural dynamic patterns of flow and propose the Global Dynamic Flow Model that characterizes flow as a high-dimensional, global metastable neural activity. Our study offers novel insights into the role of high-dimensionality and global metastability in brain activity associated with the flow experience.

1. Introduction

The study of psychological flow, which is an optimal experience, is gaining increasing attention in behavioral and neuroscience research. Flow is characterized by profound immersion, intense absorption, and enhanced cognitive functioning in carrying out an activity of interest (Csikszentmihalyi, 1990; Nakamura and Csikszentmihalyi, 2002). Flow experience is often associated with peak performance and high contentment in human experiences in activities that include gaming, sports, education, and work (Csikszentmihalyi, 2000; Csikszentmihalyi and LeFevre, 1989; Harris et al., 2023). Although the behavioral aspects of the flow experience have been widely studied (Csikszentmihalyi, 1990; Ghani and Deshpande, 1994; Jackson and Csikszentmihalyi, 1999), the dynamic neuropsychological mechanisms underlying the flow experience remain poorly understood. Specifically, how brain functional networks are dynamically coordinated during the flow experience remains unclear.

Existing neuropsychological flow theories adopt a localizationist and static perspective, focusing on specific brain functions associated with flow (Alameda et al., 2022). Neuropsychological flow research has primarily drawn on two theoretical perspectives: the Transient Hypofrontality Hypothesis (THH) (Dietrich, 2004) and the Synchronization Theory of Flow (STF) (Weber et al., 2009). THH posits that during the flow experience, there is a temporary decrease in activity in the prefrontal cortex, a brain region associated with higher-level cognitive functions (Miller and Cohen, 2001). By downregulating the prefrontal

* Corresponding author at: University of California, Santa Barbara, USA.
E-mail address: renew@ucsb.edu (R. Weber).

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cortex, the brain can free up cognitive resources for more automatic, effortless, and efficient processing. This reduced activity or "hypofrontality" allows greater focus and immersion in the task at hand as the processing in the brain shifts from conscious, deliberate control to implicit procedural systems, resulting in effortless performance and loss of self-consciousness often experienced during flow (Dietrich, 2004). In contrast, STF posits that flow emerges from the synchronization between the reward (pre-frontal) and cognitive control (frontoparietal) networks, facilitating the emergence of self-rewarding experience during flow (Weber et al., 2009). STF posits that flow is a discrete synchronization process that enhances communication between brain networks, leading to focused attention, diminished self-consciousness, and autotelic experience (Weber and Fisher, 2020; Weber et al., 2009). While both theories have been supported by empirical evidence, they provide inconsistent explanations for the neural dynamics of the flow experience, particularly regarding the role of the prefrontal cortex. Meanwhile, the methods used to test the two theories are also different. Most EEG studies appear to offer support for the THH (Knierim et al., 2021; Núñez Castellar et al., 2019; Wolf et al., 2015; Wollseiffen et al., 2016), as do some functional Near-Infrared Spectroscopy (fNIRS) studies (Hirao, 2014). In contrast, STF appears to be primarily supported by functional Magnetic Resonance Imaging (fMRI) studies (Huskey et al., 2018a; Huskey et al., 2022; Huskey et al., 2018b; Klasen et al., 2012) with some, but fewer, EEG studies also supporting STF (Wolf et al., 2015).

Gold and Ciorciari (2021) attempted to harmonize these two divergent flow theories by introducing the Internal Model of Flow (IMF). The IMF posits that the neurobiological underpinnings of flow involve internal models forged within the cerebellum during cognitive proficiency accumulation (Gold and Ciorciari, 2021). During skill acquisition, cerebellar activity initially increases to build these internal models and later decreases as performance becomes more automatic (Imamizu et al., 2000). Repeated practice refines these models, enabling smooth, efficient, and unconscious performance, which aligns with the effortless and immersive characteristics of flow (Gold and Ciorciari, 2020, 2021).

Previous theories assume that flow patterns characterize a highly static, optimized brain state, typically linked to specific brain region or system. THH emphasizes sustained reduction in activity of the prefrontal cortex (Dietrich, 2003), whereas STF highlights the synchronization between reward and control networks (Weber et al., 2009). Similarly, Gold and Ciorciari (2021) proposed that flow emerges when well-practiced tasks are executed via implicit systems that are primarily mediated by the cerebellar cortex. However, these perspectives do not take into consideration the contemporary models of the brain as a dynamic, complex system (Kelso, 2012; Kelso and Tognoli, 2007; Kotler et al., 2022; Noble et al., 2024), where cognitive states arise from continuous transitions among flexible and transient neural configurations rather than from static equilibria.

Building on the dynamic complex systems view of brain networks, we propose the Global Dynamic Flow (GDF) model and posit that flow can be better explained from a global dynamic neural perspective. Neural dynamics refer to alterations in the functional connectivity of brain networks that evolve over time (Rabinovich et al., 2008). Information exchange and diverse connectivity patterns within a dynamic neural network are crucial for explaining the brain's efficient and flexible cognitive functioning (Barnett et al., 2020; Kelso, 2012; Pessoa, 2013). Current EEG and fMRI studies of flow have predominantly emphasized either isolated brain regions or static functional networks, which risks reinforcing an overly limited localizationist view of brain functions, potentially leading to inconsistent findings. Therefore, investigating the functional brain dynamics of flow can offer new insights into its underlying neural mechanisms.

The GDF model advances three interrelated propositions about the neural dynamics underlying flow relative to boredom and anxiety. First, flow is characterized by the *highest global functional connectivity*, reflecting broad integration across distributed brain networks. Second, flow exhibits the *highest metastability and dynamic flexibility*, reflecting the brain's capacity to transiently coordinate these functional networks. Third, flow displays the *highest dimensionality* of dynamic functional connectivity (*d*FC), reflecting the diversity of dynamic connectivity patterns during flow experience. Together, these three propositions capture different but interconnected aspects of flow and posit that flow differs from boredom and anxiety through a neural dynamic mechanism marked by stronger global integration, greater flexible coordination, and richer diversity of time-varying connectivity patterns.

Functional connectivity refers to the temporal dependency of neural activity across anatomically separate brain regions, enabling seamless transfer and integration of information across brain networks (Davison et al., 2015; Park and Friston, 2013). Global functional connectivity reflects the brain's capacity to integrate distributed information into holistic, coherent cognitive operations (Achard et al., 2006; Gong et al., 2009; Farahani et al., 2019; Luppi et al., 2019). Because flow is a highly efficient cognitive state that demands the coordinated engagement of multiple cognitive functions (Huskey et al., 2022; Gold and Ciorciari, 2020), it is supported by stronger global functional network integration, which facilitates information exchange across distributed brain regions (Anderson and Barbey, 2023; Deco et al., 2021; Khine, 2024; Nocera et al., 2014). Accordingly, we propose that flow should be characterized by broader global functional connectivity than non-flow states, leading to hypothesis one.

All else being equal, high global efficiency trades metabolic efficiency for information-transfer efficiency (Bullmore and Sporns, 2012). This phenomenon poses a conundrum: how can the brain during flow exhibit both high global connectivity and metabolic efficiency? One answer lies in network architectures such as modular and small-world organizations, which are known to optimize this tradeoff (Sporns, 2011, 2012), including during flow (Huskey et al., 2018, 2022). A complementary solution is dynamic network reconfiguration, whereby the brain flexibly adjusts its connectivity in response to task demands (Tognoli and Kelso, 2014), including during flow (Kotler et al., 2022). Critically, this solution does not require sustained global integration; rather, efficiency can be achieved when periods of high global connectivity are entered and exited transiently as task demands evolve (Deco et al., 2015, 2025; Kelso, 1995; Tognoli and Kelso, 2014). This dynamic network reconfiguration is explained by metastability: a state in which brain networks alternate flexibly between synchronization and desynchronization, balancing integration with specialization (Kelso and Tognoli, 2007; Tognoli and Kelso, 2014; Hancock et al., 2025). This dynamic balance is crucial and fundamental for flexible and efficient cognitive functioning (Hancock et al., 2025).

Flow is characterized by high control and effortless engagement (Csikszentmihalyi, 1990; Marty-Dugas and Smilek, 2019). STF attributes these cognitive features to the synchronization between the reward and cognitive control networks (Weber et al., 2009). Such synchronization enables individuals to not only be highly focused and engaged but also experience intense self-motivation and autotelicity, resulting in a sense of effortlessness in managing challenges (Weber and Fisher, 2020). From a dynamic systems perspective, static synchronization might be insufficient to support complex task demands; instead, dynamic interactions among functional brain networks provide a more robust foundation for cognitive flexibility (Brinkman et al., 2022; Hancock et al., 2025; Wijaya et al., 2025). The flexible hub theory posits that the frontoparietal network (FPN) functions as a "dynamic coordinator," rapidly reconfiguring its connectivity with other networks to meet shifting cognitive demands (Cole et al., 2013b). Coordination Dynamics Theory (CDT) further extends this view by identifying metastability as the mechanism underlying such flexibility—a transient and delicate balance between integration and segregation of functional networks (Kelso, 2012; Tognoli and Kelso, 2014).

In a metastable state, the brain networks maintain their specialized functions while flexibly coordinating with other networks in response to changing demands, allowing the system to adapt dynamically without

becoming trapped in a single, rigid state (Hancock et al., 2025; Kelso and Tognoli, 2007). This dynamic interplay of metastability enables the flexible reconfiguration of functional connectivity patterns (Roberts et al., 2019), facilitating transitions across various network typologies in response to varying task demands (Hellyer et al., 2015). Additionally, metastable dynamics may not only support transitions among connectivity configurations across functional brain networks (Hellyer et al., 2015) but also increase the diversity of time-varying dynamic connectivity patterns (Rabinovich et al., 2008). In flow, such metastable dynamics may further enrich the diversity of time-varying connectivity patterns, thereby expanding the brain's dynamic repertoire for sustained adaptive control (Rabinovich et al., 2008; Tsuda, 2001). While the second proposition indicates a high degree of metastability in global dynamics during flow, it remains unknown whether the *d*FC dimensionality is also increased, which leads to our third hypothesis.

The *d*FC dimensionality refers to the number of distinct, independent neural network connectivity patterns the brain can generate over time as situational demands evolve (Ganguli and Sompolinsky, 2012; Tozzi, 2019). From a global and dynamic perspective, flow is a high-dimensional experience characterized by a complex combination of multiple cognitive features (Gold and Ciorciari, 2020; Nakamura and Csikszentmihalyi, 2014). It requires the simultaneous orchestration of multiple cognitive processes and various brain connectivity typologies under continuously evolving task demands, enhancing cognitive control and adaptability (Nakamura and Csikszentmihalyi, 2002). Thus, higher *d*FC dimensionality during the flow experience, as presented in our third proposition, further explains the diversity of time-varying connectivity patterns through which the brain integrates and coordinates multiple cognitive processes.

We tested the above three propositions on global functional connectivity, metastability, and *d*FC dimensionality in a within-subject experimental study utilizing *Tetris*, a widely used interactive game in flow research (Keller and Bless, 2007). To investigate the neural dynamics underlying the flow experience, we employed a 64-channel EEG system. Detailed descriptions of the experimental procedures and methods are provided next.

2. Method

2.1. Subjects

Data were collected from 42 right-handed male student subjects aged 18-30. They voluntarily participated in the study for extra credits in their coursework. All subjects had normal or corrected-to-normal vision

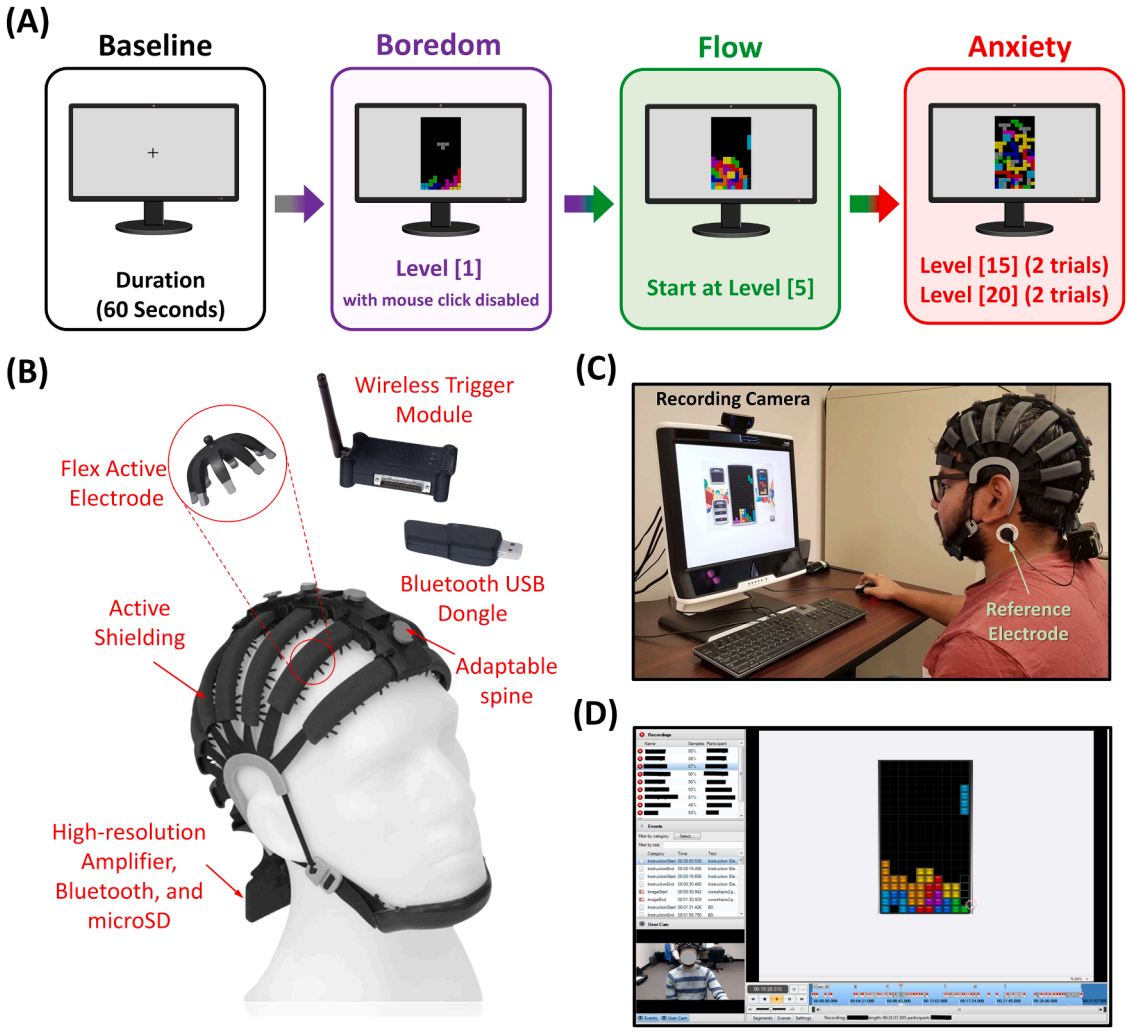


Fig. 1. The experimental design. (A) Task sessions, starting with EEG baseline collection and three linearly increasing difficulty conditions, i.e., (Boredom; difficulty < ability), (Flow; difficulty ≈ ability), and (Anxiety; difficulty > ability). (B) The Cognionics HD-72 64-channel dry-mobile wireless EEG system was used to collect brainwaves while enhancing the participant's naturalistic experience. (C) Experimental environment and setup, showing a participant playing the Tetris game while capturing brainwave oscillations and recording the participant's behaviors and reactions within the gaming session. (D) Post-experimental retrospective process tracing upon completing the gameplay to identify the time segments wherein the participant experienced boredom, flow, and anxiety.

with no history of neurological or psychiatric disorders.

2.2. Experimental design

We carried out a within-subject experimental study to induce the states of flow, boredom, and anxiety in video gaming using the Tetris game, as illustrated in Fig. 1. After the gameplay, we used the retrospective process-tracing method to prompt subjects to articulate their states of experience during the entire gameplay to determine when flow, boredom, and anxiety took place during the gameplay. Brain activity was recorded using a 64-channel EEG system, which included a Cognionics HD-72 wireless dry high-density EEG headset to capture brainwave oscillations.

Before the experiment, we explained the Tetris game to the subjects to familiarize them with the activity, after which they completed one round of gameplay as training. We selected the Tetris game as our experimental context for several reasons: (1) video games are inherently more motivating for inducing flow experiences in a laboratory setting (Gold and Ciorciari, 2019; Harmat et al., 2015; Keller and Bless, 2007; Weber et al., 2009) (2) Tetris can be used to induce the flow experience by manipulating game difficulty, and (3) Tetris is simple and familiar to many, minimizing training time.

2.2.1. Baseline condition

Before the experiment commenced, we administered a pre-study questionnaire to gather participants' demographic information. Baseline EEG signals were collected during the first 60 seconds of the session, during which participants were instructed to fixate on a cross marking in the center of a gray screen and to remain calm and relaxed without engaging in any physical activity.

2.2.2. Experimental manipulation

Based on the flow theory, the balance of challenge and skill is a necessary condition for the flow experience (Csikszentmihalyi et al., 2005). Hence, we manipulated difficulty or challenge (i.e., Tetris speed) using three different experimental conditions:

a) **Boredom (difficulty < ability):** excessively easy compared to the participant's skill level (i.e., level 1), wherein the Tetris game was consistently slow for three minutes, and the mouse click feature was disabled during gameplay to avoid participants taking shortcuts.
b) **Anxiety (difficulty > ability):** excessively difficult compared with the participant's skills, wherein the Tetris game was overly fast. Participants had two trials for each of two difficulty levels (i.e., 15 and 20) without any time limits for the gameplay.
c) **Flow (difficulty ≈ ability):** the challenge of the game and the participant's skill are roughly balanced. Based on our pilot studies, we found that most of the subjects experienced flow after level 5 of the Tetris game. Hence, we set up level 5 of the Tetris game (without specifying any time limit) for the gameplay to elicit the flow experience. The gameplay terminated when all the falling blocks were stacked up.

Finally, our experimental design started with the lowest difficulty level and gradually increased in difficulty. We did not counterbalance the conditions (after pilot testing the experiment with and without counterbalancing the conditions) because attaining the desired user experience is more difficult when the task complexity is randomized instead of growing linearly (Sangari and Zerehsaz, 2020). Participants were more likely to develop their game skills and achieve a flow experience eventually, without counterbalancing the conditions.

2.2.3. Self-reported user experience

A self-reported user experience assessment was carried out using a questionnaire to validate whether subjects experienced boredom, flow, and anxiety after each gameplay session. To assess the experimental manipulation and participants' subjective flow experience, participants first read a standardized description of flow, which characterized flow as a state of deep involvement, complete immersion, reduced awareness of irrelevant surroundings, altered sense of time, and intrinsic enjoyment. After reading this description, participants rated three Tetris-adapted flow items: "I experienced flow in the previous round of Tetris," "I felt I was in flow during the previous round of Tetris," and "In the previous round of Tetris, I experienced the flow state." Boredom and anxiety were also measured using corresponding self-report items after each session. All items were rated on a seven-point Likert scale ranging from strongly disagree to strongly agree.

To examine whether the manipulations worked as intended, we tested whether each self-reported state was highest in its corresponding session: boredom in the boredom-inducing session, flow in the flow-inducing session, and anxiety in the anxiety-inducing session. Because all three sessions were completed by each participant, we conducted the one-way repeated-measures ANOVA for each state measure, followed by paired-sample t-tests with Holm-Bonferroni correction for post hoc comparisons. The descriptive results showed clear condition-specific patterns. Boredom was rated highest in the boredom-inducing session, flow was highest in the flow-inducing session, and anxiety was highest in the anxiety-inducing session. Please refer to Supplementary Information S1 for more detailed descriptive results.

2.2.4. Post-experimental retrospective process tracing

The participant's behaviors and reactions were observed and recorded during each session. Upon completing the experimental session involving the three conditions, we carried out retrospective process tracing with the participants, wherein they watched the video recording of their gameplay session and were asked to verbalize their feelings and experiences as they were playing the game. During retrospective process tracing, we identified the time segments where participants specifically indicated that they experienced boredom, flow, and anxiety during the gameplay, e.g., 123 ± 30, 176 ± 89, and 162 ± 92 seconds, respectively.

2.3. EEG data preprocessing

The preprocessing and analysis of EEG data were conducted using the FieldTrip toolbox (Oostenveld et al., 2011). Additional custom analyses were implemented using MATLAB scripts, tailored specifically to address the dynamic connectivity analysis in this study.

Preprocessing EEG signals involved multiple steps to improve data quality and extract relevant features, as illustrated in Fig. 2A. Our proposed preprocessing pipeline followed the approach outlined in a previous EEG signal preprocessing study (de Cheveigné and Arzounian, 2018).

The preprocessing pipeline began with filtering, where EEG channels were demeaned and band-pass filtered between 0.5 Hz and 120 Hz. Power-line interference at 60 Hz was then removed using a Discrete Fourier transform (DFT) filter. To improve processing efficiency while preserving signal integrity, EEG signals were downsampled to 300 Hz.

Next, segments (epochs) of interest were identified based on retrospective process tracing and extracted for further cleaning and analysis. Noisy EEG channels with significant distortions were corrected through interpolation using neighboring channels. The EEG signals were then manually inspected to remove segments containing muscle and jump artifacts. Finally, any remaining outlier segments were automatically removed using Z-value and variance-based detection metrics when their deviation exceeded three standard deviations from the global signal variation.

Linear trends in EEG data were removed after excluding the EEG segments containing artifacts. Common Average Re-referencing (CAR) was utilized (Luck, 2014; Nunez and Srinivasan, 2006), wherein each electrode was re-referenced by subtracting the average of all electrodes' signals. Residual artifactual components, particularly those related to

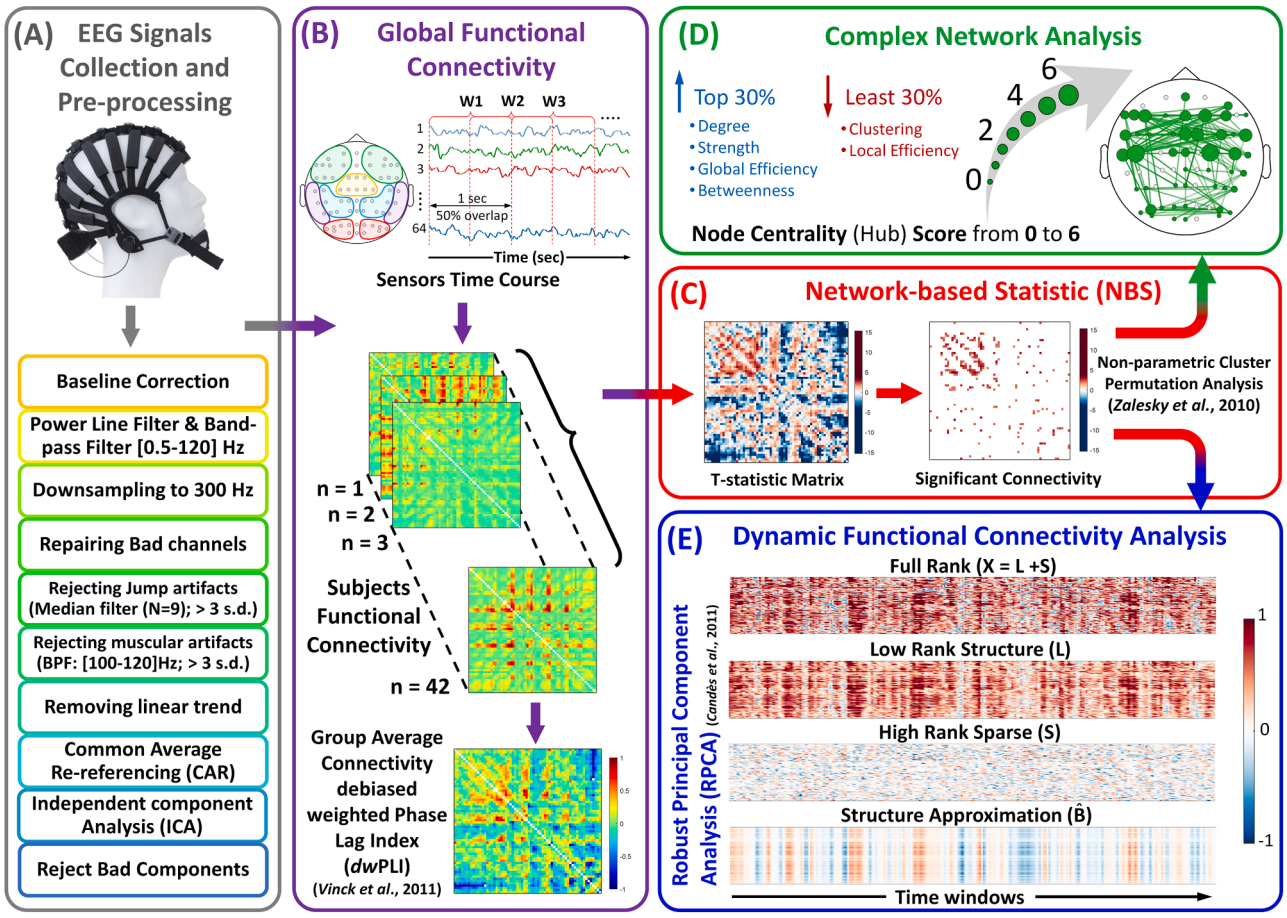


Fig. 2. Functional connectivity construction and analysis pipeline. (A) First, Preprocessing EEG signals involves several steps to enhance the quality of the data and extract relevant features. (B) Second, the group functional connectivity matrix was calculated using the debiased weighted phase lag index (*dw*PLI) after the segmentation of clean EEG data into 1-second epochs with a 50% overlap. (C) Third, the statistical inference of the observed connectivity patterns was assessed utilizing Network-Based Statistics (NBS) cluster-level permutation analysis, highlighting all significant connectivity clusters. (D) Fourth, the group representative function network was constructed after node centrality analysis, showing hubs and backbone connections. (E) Finally, the dynamic alterations of functional connectivity across time were analyzed using robust principal component analysis (RPCA) to decompose a temporal connectivity data matrix **X** into a structured low-rank matrix **L** and a sparse matrix **S** containing outliers and noise data.

eye blinks, muscle activity, and cardiac activity, were identified and excluded using independent component analysis (ICA) decomposition. Finally, the 'CleanLine' algorithm (Mitra and Bokil, 2007) was applied to adaptively estimate and remove residual sinusoidal power-line noise artifacts from the remaining ICA components. Following preprocessing, the durations of the experiences of interest dropped as follows: e.g., 114 ± 30, 170 ± 88, and 151 ± 90 in seconds for boredom, flow, and anxiety, respectively.

2.4. Global functional connectivity analysis

The process began with segmenting the preprocessed EEG data into 1-second epochs with a 50% overlap. This approach was designed to enhance temporal resolution and improve the coherence between consecutive epochs, thereby increasing the robustness of connectivity estimates. Following segmentation, the Debiased Weighted Phase Lag Index (*dw*PLI) was calculated to measure the phase synchronization (Fig. 2B), assessing the consistency of phase differences between EEG signals while minimizing the influence of noise and volume conduction (Vinck et al., 2011).

To assess the statistical significance of the observed connectivity patterns, we applied a Network-Based Statistic (NBS) cluster-level permutation test (Zalesky et al., 2010). The cluster-level permutation test utilized the Monte Carlo method with 5000 permutations, a statistic threshold p-value < 0.025, and a cluster p-value of 0.05 to control the multiple comparisons and false-positive connections influence. NBS generated a T-statistic matrix for each possible connection, as shown in Fig. 2C, and then identified clusters of connections that show statistically significant differences, thereby enhancing the robustness and reliability of the connectivity results. All the insignificant connections were then discarded from the analysis, as depicted in Fig. 2C. The functional connectivity was estimated for various EEG frequency bands (i.e., Delta, Theta, Alpha, low Beta, medium Beta, high Beta, and Gamma bands). Please refer to Supplementary information S2 for more details.

2.5. Complex network analysis

We utilized a subset of network measures of integration and segregation in human brain networks to gauge the centrality of specific recoding regions and to describe regional functional circuitry patterns: degree, strength, betweenness, global efficiency, clustering coefficients, and local efficiency (Rubinov and Sporns, 2010). Based on the hub definition established in prior research on human brain networks, hub nodes exhibit multiple topological centrality features (van den Heuvel et al., 2010).

Our study adopted this approach to assign each node a centrality score ranging from 0 to 6. The assessment began at 0, increasing by one each time a node met one of the following criteria: ranking in the top

30% for degree, strength, betweenness, and global efficiency, as well as in the bottom 30% for clustering coefficient and local efficiency. A node with a score of 6 satisfies all the proposed hub criteria. These metrics were integrated to provide a comprehensive depiction of functional connectivity across different cognitive experiences, with node size proportional to its hub score, as shown in Fig. 2D. Complex brain-network analysis was carried out using the Brain Connectivity Toolbox (Rubinov and Sporns, 2010). Please refer to Supplementary information S3 for more details.

2.6. Dynamic functional connectivity analysis

To evaluate the dynamic alterations of functional connectivity and examine the synchronization patterns across time between different experiences, we employed Principal Component Analysis (PCA) to quantify the number of distinct and independent *d*FC patterns (Preti et al., 2017). In many natural systems, data often exhibit dominant patterns that can be characterized by a low-dimensional attractor (Holmes et al., 2012). However, these dominant patterns are typically interspersed with outliers and corrupted data, which can falsely increase the observed dimensionality of the system.

The PCA is highly susceptible to outliers and noisy data. To overcome this sensitivity issue, we first utilized a robust principal component analysis (RPCA) algorithm (Candès et al., 2011). The RPCA seeks to decompose a temporal connectivity data matrix $\mathbf{X}$ into a structured low-rank matrix $\mathbf{L}$ and a sparse matrix $\mathbf{S}$ containing outliers and noise data, as presented in Fig. 2E. The principal components of the temporal connectivity structure matrix $\mathbf{L}$ are robust to the outliers after isolating the sparse matrix $\mathbf{S}$. (see Supplementary information S4 for more details)

To get the fundamental independent components that can capture the hidden pattern of the dynamic connectivity patterns, we performed Singular Value Decomposition (SVD) by the following steps: First, $\mathbf{L}$ was demeaned and normalized to neutralize the effect of observation numbers. Second, the normalized mean-subtracted dynamic connectivity matrix $\mathbf{B}$ was decomposed to $\mathbf{B} = \mathbf{U\Sigma V}^{T,}$ where the left singular vectors $\mathbf{U} \in \mathbf{R}^{m\times m}$, right singular vectors $\mathbf{V} \in \mathbf{R}^{n\times n}$, and singular values matrix $\mathbf{\Sigma} \in \mathbf{R}^{m\times n}$. Matrix $\mathbf{V}$ captures information about the dynamic principal connectivity patterns. Third, the optimal rank-r approximation matrix $\hat{\mathbf{B}}$ was obtained by keeping the leading r singular values and vectors that represent 95% of the total explained variance of matrix $\mathbf{B}$ and discarding the least 5% components that represent residuals. The $\hat{\mathbf{B}}$ connectivity matrix was utilized to estimate various parameters to assess the dynamic connectivity as follows.

2.6.1. dFC strength

We employed the Root Mean Square (RMS) measure to estimate the strength of connectivity alterations over time in a network. A high RMS of dFC indicates that the connectivity strength between different regions or nodes of a network is strong on average, suggesting consistent synchronization between network elements over time. The RMS was calculated for the *dw*PLI time series of each significant connection. Then, the average RMS value was calculated across all connections for each participant.

2.6.2. dFC flexibility and metastability

The flexibility of the connectivity dynamics can be estimated by measuring the speed of the functional connectivity fluctuations over time (Cavanna et al., 2018). The Phase Difference Derivative (PDD) has been suggested by researchers for identifying instances of phase desynchronization in high-dimensional systems (Breakspear et al., 2004). The PDD captures the speed of synch-desynch alterations over time by calculating the derivative of the phase difference. Therefore, a high absolute value of the mean PDD indicates more rapid and flexible toggling between integration and segregation experiences. Furthermore, the variance of the PDD was proposed as a valid signature of metastability in high-dimensional systems (Hancock et al., 2025; Zhang et al., 2018).

Since we constructed the functional network of different experiences based on the phase difference between brain regions, we estimated PDD as the first derivative of the time-series synchronization pattern of each functional link. Then, the average of PDD absolute and PDD variance values were computed across all significant connections and averaged for each participant to estimate link-level flexibility and metastability, respectively.

2.6.3. dFC dimensionality

To estimate the dimensionality of *d*FC, 95% of the total variance explained by each principal component of the dynamic connectivity matrix was retained across the experiences. This process followed the removal of high-rank sparse noise and excluded the least 5% of components that represent residuals. The Structure Connectivity Rank (SCR), i.e., the number of independent connectivity patterns that evolve across time, was then utilized to estimate the dimensionality of the reconstructed subspace of temporal dynamic connectivity for each state. We considered the rank (r) of approximation of the structured dynamic connectivity matrix $\hat{\mathbf{B}}$ as an estimate of the dimensionality of FC dynamics.

2.7. Statistical analysis

The statistical inference of the observed alterations in *d*FC measures across the experiences was assessed using the repeated measures one-way analysis of variance (ANOVA). All multiple comparisons were corrected by *post hoc* Tukey-Kramer's tests. All significance tests were two-tailed, and all data were presented as Mean ± Standard Deviation (SD) unless otherwise stated.

2.8. Ethics

All procedures associated with the present study were approved by the Institutional Review Board of the Missouri University of Science and Technology (Approval No.: 2017020901X) and were carried out in accordance with the guidelines and relevant ethical regulations. All participants provided written informed consent before taking part in the experiment. Participation was voluntary, and students received course credits as compensation for their involvement.

3. Results

3.1. Highest global functional connectivity during flow

We assessed global functional connectivity by quantifying phase lag consistency between every pair of EEG electrodes and averaged across all subjects. Phase consistency is a phenomenon wherein the phase angles of neural oscillations from different brain regions are correlated or synchronized over time (Fell and Axmacher, 2011). It reflects the extent to which two nodes maintain a consistent phase relationship, indicating a temporal synchrony in their oscillatory activity (Stam et al., 2007). The *dw*PLI metric was employed to estimate the consistency of the phase differences between EEG signals from all electrodes due to its robustness in mitigating noise and volume conduction (Vinck et al., 2011). Moreover, the Network-Based Statistics methodology was applied to assess the statistical significance of the observed connectivity patterns as described by Zalesky et al. (2010).

We identified the global functional networks associated with flow, boredom, and anxiety across the frequency bands. The analysis presented in Table 1 shows the normalized number of significant phase-consistent connections across the frequency bands of Delta, Theta, Alpha, low Beta, mid Beta, high Beta, and Gamma for the three experiences. Each functional connection indicates a significantly consistent phase difference between a pair of electrodes across all subjects.

Flow exhibits greater global functional connectivity in the Theta and

Table 1
Number of significant functional connections for each state across frequency bands[†].

	Delta (0.5-4 Hz)	Theta (4-8 Hz)	Alpha (8-12 Hz)	$Beta_{(l)}$ (12-15 Hz)	$Beta_{(m)}$ (15-20 Hz)	$Beta_{(h)}$ (20-30 Hz)	Gamma (30-50 Hz)
Boredom	2	9	24	43	5	0	0
Flow	3	45	122	44	6	2	0
Anxiety	0	23	54	42	28	0	0

[†] **Condition: 1 sec EEG epochs, 50% overlapping, $n = 42$, Significant functional connections were identified using cluster-based analysis with a threshold of $p < 0.025$ and a cluster-level significance of $p < 0.05$.**

Alpha bands as compared to boredom and anxiety experiences. In the Theta band, synchronization was restricted to the occipital region in the boredom and anxiety experiences but extended to the frontal region in the flow experience (Figs. 3A, 3B, and 3C). In the Alpha band, synchronization occurs in the frontal and parietal regions for all three experiences (Figs. 4A, 4B, and 4C). No significant differences were observed across the three experiences in the low Beta band (Figs. 5A, 5B, and 5C). We conducted a sensitivity analysis (see SI Fig. 6) to assess how the segment length and the degree of overlap between adjacent segments affect the global connectivity measurement outcomes. While increasing segment length or reducing overlap raised the global connectivity outcome, the relative differences across the experiences and frequency bands remained consistent, especially for Theta and Alpha bands with 25% and 50% overlap.

The flow experience has a significantly higher average network connectivity link strength (measured in Root Mean Square, RMS) across brain regions than the boredom experience in the Theta band ($F_{stat} = 56.18$, $p < 0.001$; Fig. 3D). Additionally, the flow not only involves an expanded brain functional network compared to the boredom and anxiety experiences but also exhibits significantly stronger connectivity among brain regions than the other two experiences in the Alpha band ($F_{stat} = 45.93$, $p < 0.001$; Fig. 4D). No significant differences were observed among the three experiences in the low Beta band (Fig. 5D).

3.2. Highest dFC flexibility and metastability during flow

The flexibility of connectivity dynamics can be estimated by measuring the speed of the functional connectivity fluctuations over time (Uehara et al., 2023). One such metric is the absolute Phase Difference Derivative (PDD) value, which quantifies how rapidly the phase relationship shifts between all pairs of nodes in high-dimensional systems (Breakspear et al., 2004). In neuroscience, the absolute PDD has been widely used to examine dynamic brain networks and to capture the flexible nature of functional connectivity (Tewarie et al., 2019). In the present study, we further employed the temporal variance of the PDD as an indicator of temporal variability in phase alignment, thereby serving as an index of metastability (Hancock et al., 2025). Based on the global functional connectivity results, we examined the Theta (4–8 Hz), Alpha (8–12 Hz), and low Beta (12–15 Hz) bands for the dynamic connectivity analysis, as the other frequency bands did not show significant global connectivity for the flow experience.

In the Theta band, the flow and anxiety experiences have greater absolute PDD values than the boredom experience ($F_{stat} = 51.36$, $p < 0.001$; Fig. 3E). Moreover, in the Alpha band, the flow experience exhibited a significantly higher absolute PDD value than the boredom and anxiety experiences ($F_{stat} = 44.47$, $p < 0.001$; Fig. 4E), indicating more rapid on-off synchronization switching and greater network flexibility.

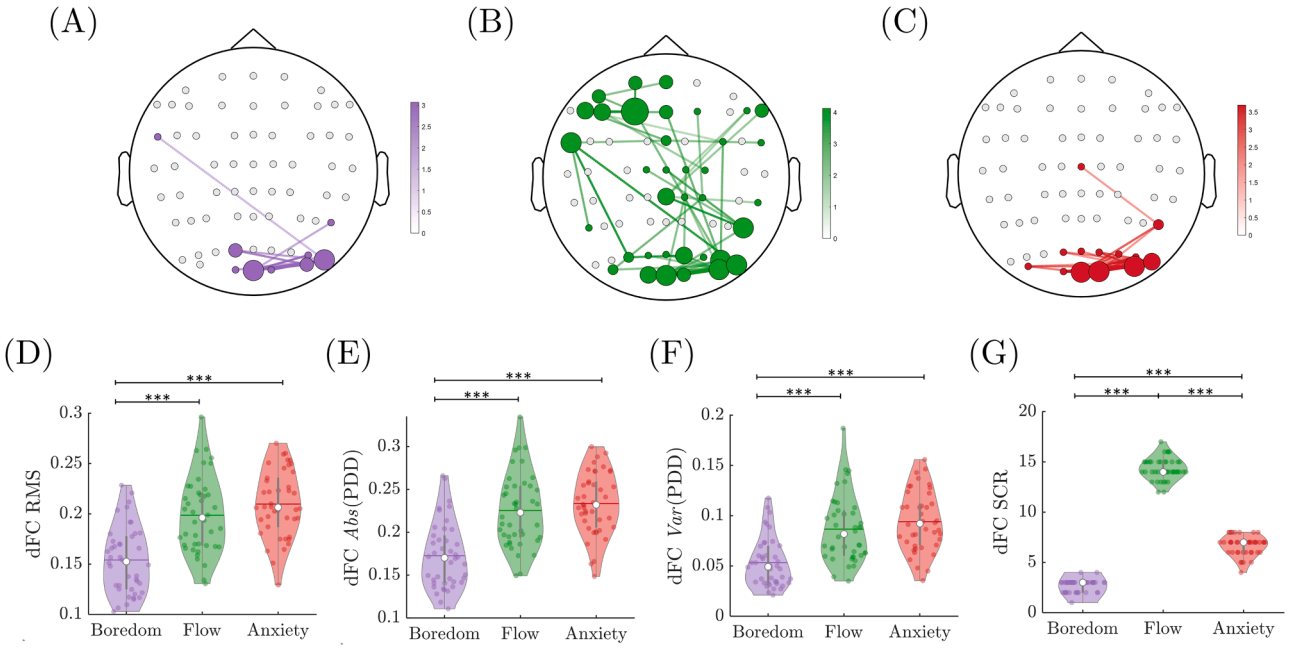


Fig. 3. Functional connectivity analysis for the Theta band. The brain's global functional connectivity network for boredom (A), flow (B), and anxiety (C) that demonstrates the spatial distribution of the dominant nodes and connections within the identified functional network. Node size indicates its hub score after centrality analysis, and the color line between nodes denotes the *T*-stat of the significant connections ($n = 42$). Significant functional connections were identified using cluster-based analysis ($p < 0.025$, cluster-level $p < 0.05$). The *d*FC analysis across the three experiences indicates the temporal activity of the identified functional networks: Root-Mean Square (RMS) (D), absolute (E) and variance (F) of Phase Difference Differential (PDD), and Structure Connectivity Rank (SCR) (G). In the violin plot, scatter points display the individual data point values, the horizontal line represents the mean, the white circle indicates the median, and the bar around the median denotes the interquartile range (IQR, q_3-q_1). All multiple comparisons are corrected using the *post hoc* Tukey tests (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

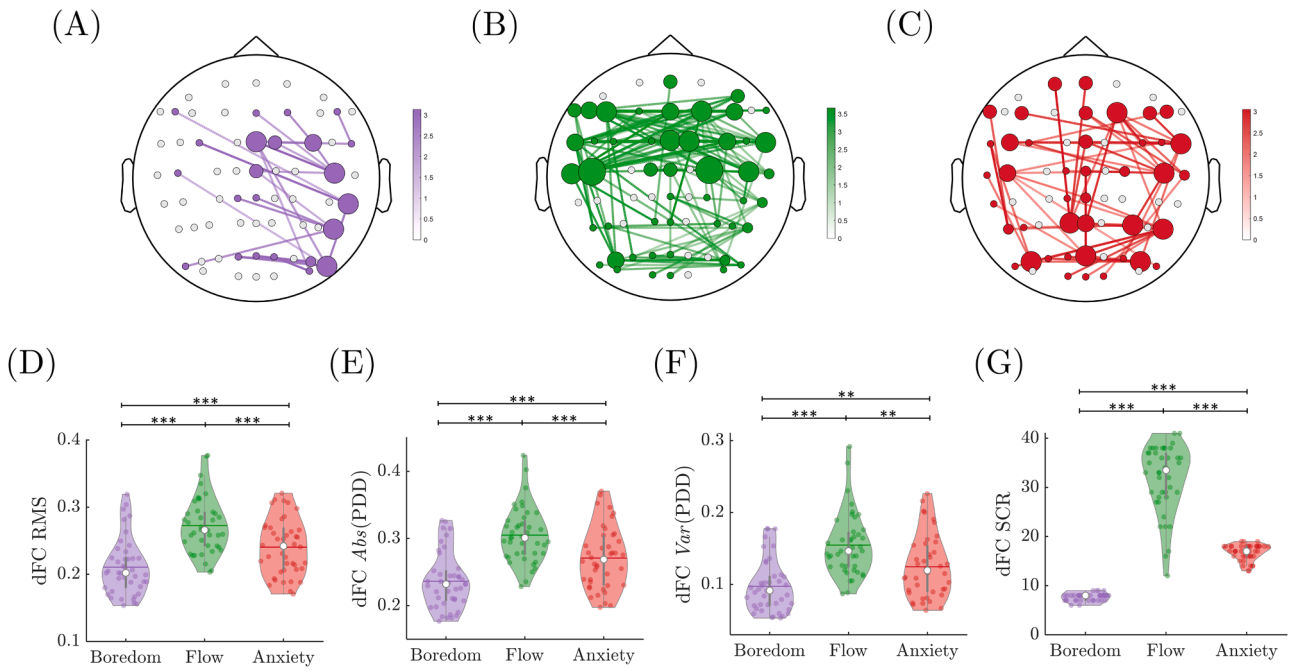


Fig. 4. Functional connectivity analysis for the Alpha band. The brain's global functional connectivity network for boredom (A), flow (B), and anxiety (C) that demonstrates the spatial distribution of the dominant nodes and connections within the identified functional network. Node size indicates its hub score after centrality analysis, and the color line between nodes denotes the *T*-stat of the significant connections ($n = 42$). Significant functional connections were identified using cluster-based analysis ($p < 0.025$, cluster-level $p < 0.05$). The dFC analysis across the three experiences indicates the temporal activity of the identified functional networks: Root-Mean Square (RMS) (D), absolute (E) and variance (F) of Phase Difference Differential (PDD), and Structure Connectivity Rank (SCR) (G). In the violin plot, scatter points display the individual data point values, the horizontal line represents the mean, the white circle indicates the median, and the bar around the median denotes the interquartile range (IQR, q_3-q_1). All multiple comparisons are corrected using the *post hoc* Tukey tests (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

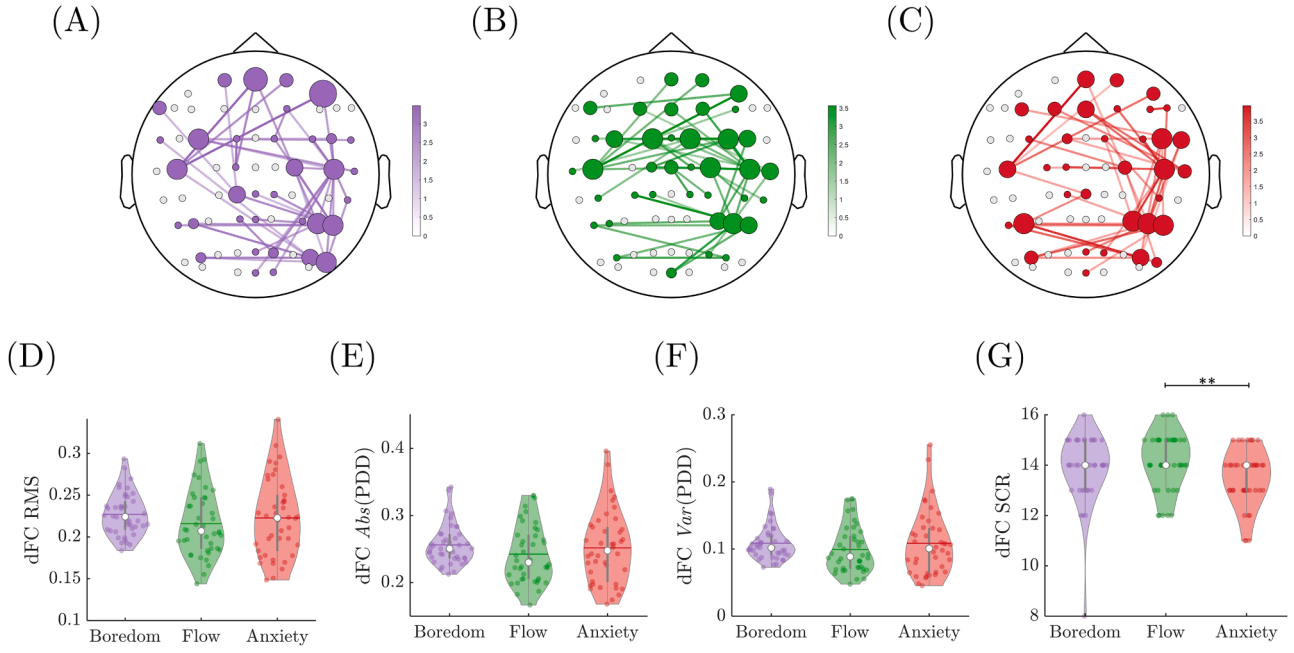


Fig. 5. Functional connectivity analysis for the low Beta band. The brain's global functional connectivity network for boredom (A), flow (B), and anxiety (C) that demonstrates the spatial distribution of the dominant nodes and connections within the identified functional network. Node size indicates its hub score after centrality analysis, and the color line between nodes denotes the *T*-stat of the significant connections ($n = 42$). Significant functional connections were identified using cluster-based analysis ($p < 0.025$, cluster-level $p < 0.05$). The *d*FC analysis across the three experiences indicates the temporal activity of the identified functional networks: Root-Mean Square (RMS) (D), absolute (E) and variance (F) of Phase Difference Differential (PDD), and Structure Connectivity Rank (SCR) (G). In the violin plot, scatter points display the individual data point values, the horizontal line represents the mean, the white circle indicates the median, and the bar around the median denotes the interquartile range (IQR, q_3-q_1). All multiple comparisons are corrected using the *post hoc* Tukey tests (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Additionally, the flow experience showed a significantly higher variance of PDD values than the boredom in the Theta band (F_{stat} =

40.19, $p < 0.001$; Fig. 3F) and a significantly higher variance of PDD values than the other two experiences in the Alpha band ($F_{stat} = 37.33$, $p < 0.001$; Fig. 4F). The anxiety experience also exhibited significantly higher value in both the absolute and variance of PDD values compared to the boredom in the Theta band (Abs (PDD): $p < 0.001$; Var (PDD): $p < 0.001$; Figs. 3E and 3F, respectively) and the Alpha band (Abs (PDD): $p < 0.001$; Var (PDD): $p < 0.001$; Figs. 4E and 4F, respectively). No significant difference in connectivity dynamics of the Theta band was observed between the flow and anxiety experiences (Abs (PDD): $p = 0.432$; Var (PDD): $p = 0.332$; Figs. 3E and 3F, respectively). This could be due to the rapid pace of visual stimulation associated with the challenge posed by the experimental design of anxiety experience. Moreover, there is no significant difference in the absolute and variance of PDD values across the three experiences in the low Beta band (Abs (PDD): $F_{stat} = 1.77$, $p = 0.177$; Var (PDD): $F_{stat} = 1.34$, $p = 0.276$; Figs. 5E and 5F, respectively).

3.3. Highest dFC dimensionality during flow

We employed the SCR of the identified functional network dynamics as a measure for dFC dimensionality. Among all three experiences, the flow experience exhibited a significantly higher connectivity rank in both Theta ($F_{stat} = 2340.48$, $p < 0.001$; Fig. 3G) and Alpha ($F_{stat} = 398.09$, $p < 0.001$; Fig. 4G) bands than the other two experiences, and hence, the flow experience has the highest dFC dimensionality. In the low Beta band, the flow experience showed a higher SCR than the anxiety experience ($p < 0.01$; Fig. 5G). On the other hand, anxiety demonstrated a significantly higher SCR value of the dFC compared to boredom in the Theta ($p < 0.001$) and Alpha ($p < 0.001$) bands, but not in the low Beta band ($p = 0.237$). These findings highlight the diversity of distinct dynamic connection patterns between different brain regions during flow. All the detailed statistical results are shown in Table 2.

4. Discussion

To advance our understanding of the neural dynamics underlying the flow experience, we propose the Global Dynamic Flow (GDF) Model, which extends current theories by modeling flow as emerging from global, metastable, and high-dimensional brain network dynamics. Using EEG-based global and dynamic functional connectivity analyses, our research demonstrates that the flow experience is characterized by broader and more robust global neural connectivity, suggesting more efficient information transmissions during flow than non-flow experiences—boredom and anxiety. Furthermore, the flow experience shows greater metastability and higher dimensionality than anxiety and boredom experiences, suggesting a more flexible and adaptive brain network configuration. These dynamic properties provide critical insight into how the brain efficiently coordinates multiple cognitive functions and rapidly adapts to varying task demands during flow. Such flexible coordination facilitates a a perceived sense of control and effortless engagement that are hallmarks of the flow experience (Csikszentmihalyi, 1990). Importantly, these results offer a novel perspective by indicating that flow should not only be conceptualized as merely the activation or deactivation of a static, localized brain region or network, as implied by earlier flow theories (Dietrich, 2003; Gold and Ciorciari, 2020; Weber et al., 2009). Instead, our findings suggest that the flow experience emerges from a dynamic, global interplay among multiple brain networks, showing a higher level of metastability within the brain's functional networks.

Our results indicate that the flow experience is characterized by higher and more robust brain global connectivity in the Theta and Alpha bands. These results suggest that, in contrast to boredom and anxiety, flow involves more brain functional networks during the neural dynamic process. The brain functions as a complex system, and its intricate functions emerge due to the dynamic cooperation between individualized neuronal ensembles (Kelso, 1995; Yuste et al., 2024). Enhanced brain-wide functional connectivity represents a fundamental neural signature of enhanced cognitive functions (Farahani et al., 2019). This large-scale integration manifests as increased functional connectivity between distant cortical regions, particularly involving FPN, which serve as convergence zones for multimodal information (Cole et al., 2013b; Crossley et al., 2013). During conscious processing, sensory information propagates beyond primary cortical areas to engage association cortices, pre-frontal regions, and subcortical structures, creating a "global workspace" where information becomes available for flexible, goal-directed operations (Deco et al., 2021).

The findings align with the perspective of the STF (Weber et al., 2009), which posits that the flow experience arises from the synchronization of various brain networks, particularly those involved in cognitive control and reward processing. Supporting this notion, Huskey et al. (2022) also demonstrated increased global brain connectivity during flow. This enhanced functional integration explains why individuals report a profound sense of effortless control during flow experience. The strengthened direct connections between nodes and the expanded functional connectivity enable the brain to meet diverse task demands effectively while maintaining the subjective sense of mastery and seamless engagement that define the flow experience (Cocchi et al., 2013).

Furthermore, these results also align with the notion of Compositional Coding from the Flexible Hub Theory in explaining rapid knowledge transfer and mental flexibility (Cole et al., 2013b). Compositional Coding suggests that rather than being constructed from scratch, the integrative process between the pre-frontal cortex and other brain networks builds on prior experiences through a form of neural transfer learning (Cole et al., 2011). This mechanism explains the remarkable human ability to rapidly apply cognitive control to new tasks via the reuse of task elements based on old task contexts (Cole et al., 2013a). Specifically, the cognitive control process would allow neural activity

Table 2
Comparisons between flow, boredom, and anxiety experiences[†].

	Metrics	$F_{(2,82)}$-static	*P*-value	Boredom (B)	Flow (F)	Anxiety (A)	F − A	F − B	A − B
Theta	**RMS**	56.18	**<0.001**	0.154±0.033	0.199±0.038	0.209±0.033	0.126	**<0.001**	**<0.001**
	Abs(PDD)	51.36	**<0.001**	0.173±0.038	0.226±0.043	0.234±0.036	0.432	**<0.001**	**<0.001**
	Var(PDD)	40.19	**<0.001**	0.053±0.025	0.087±0.033	0.094±0.029	0.332	**<0.001**	**<0.001**
	SCR	2340.48	**<0.001**	2.691±0.604	14.309±1.07	6.691±0.975	**<0.001**	**<0.001**	**<0.001**
Alpha	**RMS**	45.93	**<0.001**	0.211±0.042	0.272±0.041	0.240±0.042	**<0.001**	**<0.001**	**<0.001**
	Abs(PDD)	44.47	**<0.001**	0.236±0.042	0.305±0.041	0.271±0.046	**<0.001**	**<0.001**	**<0.001**
	Var(PDD)	37.33	**<0.001**	0.097±0.036	0.155±0.044	0.125±0.043	**<0.01**	**<0.001**	**<0.001**
	SCR	398.09	**<0.001**	7.714±0.708	31.762±6.98	16.786±1.631	**<0.001**	**<0.001**	**<0.001**
$Beta_{(l)}$	**RMS**	1.06	0.207	0.227±0.026	0.216±0.039	0.222±0.046	0.5490	0.2004	0.7466
	Abs(PDD)	1.77	0.177	0.256±0.029	0.242±0.043	0.252±0.054	0.4753	0.1580	0.8087
	Var(PDD)	1.34	0.267	0.109±0.026	0.099±0.035	0.109±0.047	0.3782	0.2961	0.9999
	SCR	4.77	**<0.05**	13.905±1.322	14.143±1.117	13.524±1.11	**<0.01**	0.5065	0.2371

[†] A repeated measure one-way ANOVA test was applied to assess the statistical significance of observed changes in *d*FC across the experiences. All multiple comparisons are corrected by *Tukey-Kramer's* tests. Data are presented as (Mean ± Standard Deviation).

and connectivity patterns that are refined and strengthened during previous cognitive activities to be reused in novel task contexts by integrating with new functional networks, facilitating the transfer of task knowledge and skill to handle new challenges (Cole et al., 2011). Thus, this "refine and reuse" process could facilitate the flexible cognitive control during the flow experience, explaining the expanded brain network in synchrony and strong node connections in the brain's dynamic connectivity during flow.

Importantly, this broader global integration need not imply sustained, metabolically costly whole-brain coupling. Instead, our dynamic results are consistent with the idea that flow realizes high communication efficiency through metastable coordination, briefly entering and exiting more integrated configurations as task demands fluctuate, thereby preserving flexibility while limiting the energetic burden of tonic global integration (Kotler et al., 2022). The results demonstrate the brain's enhanced capacity to transiently toggle between integration and segregation (Tognoli and Kelso, 2014), balancing coordination among functional networks in response to the external environment task demands during flow (Huskey et al., 2022; Kotler et al., 2022). These findings contribute to flow theories by demonstrating that flow represents a highly dynamic metastable process, rather than a stable or optimized brain state (Dietrich, 2003; Gold and Ciorciari, 2020; Weber et al., 2009).

Our metastable view of flow experience offers a neuroscientific explanation for the Information Theory of Flow (ITF), which posits that flow is an increasing function of the mutual information, denoted as $I(M;E)$, between desired end experiences (E) and means of attaining them (M) (Melnikoff et al., 2022). This mechanism helps restrict learning efforts to valid action-outcome pairs, avoiding the costly pursuit of meaningless associations. When $I(M;E)$ is high, learning promotes immersion (i.e., flow) to facilitate learning; when $I(M;E)$ is low, it encourages disengagement (e.g., boredom) to conserve cognitive resources (Melnikoff et al., 2022). Consistent with the result that the brain attentional network exhibits non-linear criticality during flow (Weber et al., 2018), our study demonstrates that the brain exhibits the highest metastable dynamics during flow. During flow, the metastability dynamic enables functional brain networks to reconfigure flexibly through dynamic transitions between brain functional network synchrony and desynchrony in response to task-related feedback represented by $I(M;E)$. This flexible reconfiguration supports efficient learning by allocating cognitive resources toward valid action–outcome contingencies, as reflected in increased brain synchrony, while simultaneously promoting brain desynchrony to prevent the costly pursuit of unproductive associations.

Our findings further indicate that the flow experience is associated with a higher dimensionality of brain network connectivity configurations than boredom and anxiety experiences, which represents more diverse dynamic connectivity patterns during the flow. This finding advances our understanding of the flexible cognitive control that characterizes the flow experience (Moore, 2013; Nakamura and Csikszentmihalyi, 2014), demonstrating that higher metastability during flow not only facilitates more flexible transitions among different brain functional network topologies, but also promotes the emergence of a more diverse array of *d*FC patterns (Tsuda, 2001) compared to boredom and anxiety. From the dynamic interaction perspective, the flexible transitions and diverse *d*FC patterns collectively contribute to the brain's functional diversity (Tozzi, 2019). Since brain functional diversity supports and forms the foundation of adaptive cognitive control (Bassett and Bullmore, 2006; Miyake et al., 2000), the high-dimensionality of flow's dynamics explains how people simultaneously accommodate the demands of complex tasks and the need for flexible control. This, in turn, contributes to the subjective experience of effortlessness that is the main characteristic of flow (Csikszentmihalyi, 1990; Davison et al., 2015).

Notably, our findings show that the flow experience is characterized by the highest metastable dynamics, dimensionality, and brain network connectivity in the Alpha and Theta bands, compared to boredom and anxiety. Consistent with previous research, Theta and Alpha oscillations emerged as the dominant components during the flow experience (Katahira et al., 2018; Rácz et al., 2025) and play distinct yet complementary roles in cognitive control (Pscherer et al., 2023).

Theta oscillations are thought to be involved in working memory (Colgin, 2016; Harris and Gordon, 2015; Korotkova et al., 2018; Roux and Uhlhaas, 2014). Research demonstrated the association of Theta band connectivity in successful declarative memory encoding and retrieval in adults (Backus et al., 2016; Kaplan et al., 2014) and children (Gander et al., 2025; Johnson et al., 2022). Long-range Theta synchronization facilitates the transfer of information between the hippocampus and neocortical regions during episodic memory encoding and consolidation (Buzsáki and Moser, 2013; Colgin, 2013). This theta-mediated connectivity extends beyond memory systems to support executive functions during cognitive control and error prediction (Cavanagh and Frank, 2014), with tasks designed to involve multiple brain regions such as cross-sensory object recognition (von Stein and Sarnthein, 2000). Theta coherence between frontal and parietal brain regions, i.e., FPN, was found to be increased with memory load using a visuospatial task (Payne and Kounios, 2009; Sauseng et al., 2005). Additionally, Sauseng et al. (2007) demonstrated that interregional connectivity in the Theta oscillations was modulated by memory-related executive functions independently of task difficulty. The interregional synchronization of theta oscillations organizes working memory subprocesses, integrating between sensory bottom-up and top-down memory representations (Roux and Uhlhaas, 2014; Sauseng et al., 2010). Thus, these results indicate that Theta global connectivity plays a significant role in integrating sensory information into central executive memory processes.

Our results indicate that in the Theta band, the flow experience shows unique neural dynamics that reflect metastable coordination between the frontal, parietal, and occipital cortex. In contrast, the other two states exhibit Theta connectivity only within the occipital region. The occipital lobe plays a crucial role in visual processing and perception (Palejwala et al., 2020), while FPN operates as a flexible hub organizing information flow between brain networks (Cole et al., 2013a). Strong evidence suggests that human capacity for flexible task management is primarily attributed to the FPN (Crossley et al., 2013). The FPN integrates various cognitive processes, creating a global workspace that adapts to the ever-changing cognitive demands (Deco et al., 2021). During anxiety, the challenge of coping with the game's demands contributed to heightened feelings of anxiety and impaired functioning of the FPN, which became less able to adapt flexibly to changing demands (Park and Moghaddam, 2017). Together, these findings highlight the vital role of the FPN in sustaining high performance during flow by flexibly coordinating functional networks.

Therefore, the Theta connectivity dynamics during the flow experience seem to be characterized by high-dimensional metastable coordination between the FPN and other cognitive functional networks. In contrast, the anxiety experience exhibits high metastable Theta *d*FC that is limited to visual processing (occipital cortex). As participants visually track fast-moving objects and simultaneously struggle to cope with the game's challenges, the rapid motion may place excessive demands on the visual cortex, contributing to feelings of anxiety. These suggest that the FPN impairs the flexible coordination between association brain networks in response to high-cognitive task demands. On the other hand, the low-speed game design in the boredom state may lead to low cognitive demands (Westgate and Wilson, 2018), resulting in a brain global FC confined to the occipital area. This result highlights the important role of the FPN in cognitive processes by flexibly coordinating brain functional networks to sustain high performance during flow.

Alpha rhythms have long been thought to reflect local and global neuronal synchronization associated with cognitive information processing, such as cognitive control, attention, inhibition, consciousness, intelligence, and creativity (Bazanova and Vernon, 2014; Nobukawa

et al., 2020; Palva and Palva, 2007, 2011; Sadaghiani and Kleinschmidt, 2016; Zakharov et al., 2020). Alpha-band connectivity mainly plays a dual role in cognitive processing, serving both as a mechanism for active suppression of task-irrelevant information (inhibition) and as a top-down control of sensory processing (timing), which enable controlled knowledge access and semantic orientation (Klimesch, 2012). Rather than simply reflecting cortical idling, alpha synchronization between brain regions implements functional inhibition that gates information flow according to task demands, with increased alpha connectivity suppressing task-irrelevant regions (Foxe and Snyder, 2011; Jensen and Mazaheri, 2010). Moreover, long-range alpha connectivity supports working memory maintenance by protecting information from interference, with the phase of alpha oscillations coordinating the timing of neuronal excitability across distributed cortical sites (Bonnefond and Jensen, 2012; Magosso and Borra, 2024; Palva and Palva, 2011).

In addition, the functional significance of alpha connectivity is further evidenced by its systematic modulation with cognitive load and attentional state (Crespo-Garcia et al., 2013; Lobier et al., 2018), with optimal cognitive performance often associated with a precise balance between alpha-mediated inhibition and excitation across relevant brain networks (Sadaghiani and Kleinschmidt, 2016; Weisz et al., 2011). These fluctuations in large-scale synchrony of Alpha band showed a positive correlation with the activity in the FPN, as measured by fMRI (Sadaghiani et al., 2012), which has been suggested for phasic modulation of top-down cognitive control (Zanto et al., 2011). Hence, these results denote that Alpha global connectivity is crucial for implementing cognitive flexibility by coordinating distributed brain networks and gating information flow.

We observed no significant differences in low beta-band activity among the flow, anxiety, and boredom experiences. This absence of differentiation can be attributed to the functional role of low beta oscillations that are associated with physical stillness and internal focus (Abhang et al., 2016; Marzbani et al., 2016). Because participants were instructed to minimize unnecessary movements and remain mentally engaged across all three experiences, low-beta activity likely exhibited comparable patterns across conditions, reflecting a shared attentional mechanism for processing sensory input (Egner and Gruzelier, 2001; Sazgar and Young, 2019), and thus no significant between-experience differences.

4.1. Revisiting theories of flow

The current theoretical models of flow have emphasized mechanisms such as prefrontal deactivation (THH; Dietrich, 2004), synchronization of attentional and reward networks (STF; Weber et al., 2009), and integrations of THH and STF via internal models of motor control (IMF; Gold and Ciorciari, 2021). These seemingly unrelated theories offer complementary perspectives on the flow state. The central aim of the present work is to test whether the existing neurocognitive theories of flow can accommodate the dynamic network signatures we observed. Below, we evaluate THH, STF, and IMF against our findings.

Transient Hypofrontality Hypothesis (THH). THH posits that flow arises from the downregulation of prefrontal networks, allowing more efficient, automatic processing (Dietrich, 2004). Our findings are not consistent with this view. The elevated global connectivity and metastability observed are incompatible with a simple, localized downregulation of prefrontal function. Instead, our analysis suggests that prefrontal regions remain dynamically integrated with broader brain networks in a metastable manner, rather than becoming functionally disengaged. As originally formulated, THH lacks the theoretical apparatus to explain the diverse, flexible, time-varying coordination of the brain functional network during flow. Rather than being globally suppressed, prefrontal networks appear to function as a coordinator within a dynamically reorganizing neural system during flow.

Synchronization Theory of Flow (STF). STF proposes that flow emerges from the coordinated synchronization of attentional and reward networks (Weber et al., 2009). Our findings provide substantial, though qualified, support for STF. The increased global connectivity during flow is consistent with large-scale network synchronization, and the engagement of task-positive attentional and frontoparietal systems aligns with STF's emphasis on attentional coordination. However, STF, in its standard formulation, describes a relatively static synchronization pattern and does not predict the time-varying properties we observed, specifically, higher metastability and greater *d*FC dimensionality. These findings suggest that flow is not characterized by stable synchronization but by a dynamic regime in which synchronization and desynchronization alternate in a coordinated, flexible manner.

Internal Model of Flow (IMF). IMF is attempting to integrate prefrontal and network-synchrony accounts by invoking cerebellar contributions to internal models of motor control (Gold and Ciorciari, 2021). Our findings are compatible with the IMF's integrative perspective that flow is not solely attributable to localized brain networks but rather emerges from coordinated activity across multiple distributed neural systems. Nevertheless, our results, which indicate whole-brain dynamic coordination during flow, are silent about cerebellar contributions to flow.

Toward a Global Dynamic Account of Flow. Neural localizationism attributes cognitive functions to a static, isolated structure of the brain. This perspective even extends to a certain functional brain network or system. Hence, conceptualizing a segregated, static, functional network or system as exclusively accountable for a cognitive process represents another form of localization. Thus, all three theories, despite their unprecedented contributions to understanding the neural basis of flow experience, inherit to varying degrees a narrow localizationist framing in which cognitive functions are assigned to specific regions (THH), specific networks (STF), or specific inter-system loops (IMF). While the three models are well-supported by empirical evidence, they present only fragmented parts of the whole picture, providing differing explanations for the neural dynamics underlying the flow experience. None of them, in their current formulations, can fully account for the distributed, metastable, high-dimensional network dynamics observed during flow.

Consistent with our research model, recent work shows that functionally anticorrelated networks can be simultaneously active during flow (Kotler et al., 2022; van der Linden, 2021). These include task-negative networks, such as the default mode network (DMN), which are typically associated with internally oriented and self-referential processing. They also include task-positive networks, such as the dorsal attention network (DAN) and FPN, which are typically involved in externally oriented attention and cognitive control (Kotler et al., 2022). This suggests that flow cannot be adequately explained by the activation or suppression of isolated regions or networks. Instead, it should be understood as a dynamic network phenomenon in which distributed brain systems are globally integrated, flexibly coordinated, and dynamically diverse over time.

The proposed GDF model captures the multiple, interacting cognitive processes underlying flow, offering a broader perspective than the localizationist paradigms. Our findings of increased *d*FC metastability and dimensionality during flow are inconsistent with a strict interpretation of THH and extend beyond the explicit predictions of STF and IMF. By capturing the fine-grained temporal dynamics of whole-brain activity, we reveal that flow involves a distributed, metastable orchestration of neural processes that cannot be concluded to static network configurations or regional activity patterns—a complexity that no single, localized structure or system can fully account for.

4.2. Implications and limitations

This study advances theoretical understanding of flow from the perspective of complex dynamic neural systems. We extend the static theoretical perspectives (i.e., THH, STF, and IMF) toward a GDF model

that conceptualizes flow as a symphony composed of globally coordinated time-varying neural systems. Specifically, flow is characterized as a high-dimensional metastable regime in which brain networks are globally integrated, flexibly coordinated, and multi-dimensional. This theoretical perspective moves beyond localizationist and static paradigms that are insufficient to capture the temporal complexity of flow. It also reconciles apparent contradictions across prior studies and opens new avenues for mapping individual differences in flow experience to distinct dynamical patterns of brain organization.

Our study also offers practical insights for adaptive system design in flow induction. For example, the Dynamic Difficulty Adjustment system (DDAs)—which adapts tasks in real time to match an individual's skill level (Zohaib, 2018)—could be enhanced by incorporating high-dimensional and flexible dynamic features as difficulty adjustment design principles. Increasing the dimensionality of system design features ensures that individuals are not confined to a single problem-solving path but can draw on multiple ways of combining strategies as tasks become more challenging (Dajani and Uddin, 2015; Ionescu, 2012). Flexible design further ensures that users maintain autonomy in adopting diverse strategies as task difficulty level increases (Moore, 2013; Schutte and Malouff, 2020). Our findings demonstrate that flow is underpinned by high-dimensional and flexible brain network dynamics, highlighting the importance of designing DDAs that incorporate these two features to more effectively induce the flow experience.

The present study has limitations. The precise brain regions associated with the brain activity were not pinpointed precisely. This imprecision is due to the limitations in the spatial resolution of current EEG systems. Future studies might benefit from utilizing methods with higher spatial accuracy, such as source localization techniques, to localize brain activity. In addition, although our frequency-band selection strategy helped reduce the burden of multiple testing and enabled a more logical and focused analysis, we acknowledge that the same empirical dataset was used for both band selection and subsequent hypothesis testing. This approach may limit the extent to which the frequency-specific findings can be generalized beyond the current sample. Future studies could further strengthen this line of evidence by using independent validation samples or data-splitting procedures.

5. Conclusion

The present study introduces a Global Dynamic Flow Model, which indicates that the flow experience is underpinned by enhanced brain global connectivity, the highest levels of metastable brain dynamics, and neurodynamic dimensionality compared to non-flow experiences (anxiety and boredom). These results suggest that during flow, the brain shows enhanced capacity for flexible and adaptive reconfiguration of functional networks. Such dynamic coordination serves as a fundamental mechanism that enables the brain to efficiently integrate multiple cognitive processes and adapt seamlessly to escalating task demands, providing the basis for the effortless control that characterizes the flow experience. Overall, this study advances theoretical understanding of flow through the lens of complex networks and dynamic neural systems, rather than viewing it as a localized, static brain state.

CRediT authorship contribution statement

Abdelrahman B.M. Eldaly: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kris Zhangguang Kang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fiona Fui-Hoon Nah:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Leanne Lai-Hang Chan:** Writing – review & editing, Validation, Supervision, Resources, Methodology. **Keng Leng Siau:** Supervision, Resources, Methodology, Funding acquisition, Data curation, Conceptualization. **Xiao Fan Liu:** Writing – review & editing, Validation, Resources. **Richard Huskey:** Writing – review & editing, Validation, Methodology. **Langtao Chen:** Investigation, Data curation. **Tejaswini Yelamanchili:** Methodology, Investigation, Data curation, Conceptualization. **René Weber:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2026.122049.

Data availability

The data and code that support the findings of this study will be made available upon request.

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